Project Design Phase

Problem – Solution Fit Template

Date: 15 February 2025

Team ID: LTVIP2025TMID35065

Project Name: Trafficelligence: Advanced Traffic Volume Estimation with Machine Learning

Maximum Marks: 2 Marks

Purpose:

Solve complex problems in a way that fits the state of your customers.
Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
Sharpen your communication and marketing strategy with the right triggers and messaging.
Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
Understand the existing situation in order to improve it for your target group.

Problem – Solution Fit Template

1. Target Group (Customers/Users)

Urban traffic departments, smart city planners, emergency service coordinators, public transport authorities.

2. Existing Problems (Pain Points)

- Sudden and unpredictable traffic congestion.
- Lack of accurate forecasting during special events or weather disruptions.
- Traffic signals work on fixed timers, not real-time traffic flow.
- Inefficiencies in emergency routing and public transport scheduling.

3. Current Behaviors / Alternatives

- Manual estimation using surveillance cameras.
- Basic mobile map apps showing real-time congestion.
- Traditional traffic signal systems without AI adaptation.
- Static data reports with no predictive insight.

4. Why Existing Solutions Don’t Work

- Limited to reactive measures.
- Do not incorporate weather, holidays, or temporal patterns into planning.

- High human dependency leads to error and inefficiency.
- Inability to scale and adapt to dynamic traffic flow in real time.

5. Our Proposed Solution

- "Trafficelligence" is a machine learning-based system that uses past traffic patterns, weather data, time of day, holidays, and more to predict upcoming traffic volumes.
- Multiple regression models like Linear Regression, Random Forest, XGBoost, and SVR are used for comparative performance.
- The system outputs traffic forecasts which can be used to improve signal timings and route optimization.

6. Why It Solves the Problem

- Predictive analytics allow traffic authorities to take preemptive action.
- Reduces congestion and travel delays.
- Supports better planning for public transport and emergency services.
- Increases trust in smart city traffic systems.

7. Key Metrics for Success

- Prediction accuracy (R^2 Score, RMSE, MAE).
- Percentage decrease in average travel time during peak hours.
- System adoption and satisfaction among authorities.
- Reduction in fuel wastage and carbon emissions due to idling.

References:

https://www.ideahackers.network/problem-solution-fit-canvas/

https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe